Sensitivity, Uncertainty, Information Content, and Channel Selection for Hyperspectral Reflective Solar Retrievals of Cirrus Cloud Properties

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***Abstract*— Inferring cirrus cloud properties from spaceborne observations inevitably involves uncertainties from both forward models and measurements. These uncertainties are present in the retrieval products and propagate to downstream data-ingesting applications in atmospheric and climate modeling. Satellite reflective solar hyperspectral instruments are highly information dense and offer a means to reduce uncertainties in cirrus cloud property retrieval products. However, hyperspectral instruments contain a large number of channels imposing large computational expenses both with measurement data and radiative transfer modeling needed to perform cirrus cloud property retrievals. While using all the channels will maximize the information content, the computational expense may not be acceptable. The purpose of this study is to objectively identify channel subsets in the spectral coverage of the Climate Absolute Radiance and Refractivity Observatory Pathfinder (CLARREO-PF) instrument that, when used as reflectance inputs to a cirrus radiative and microphysical property retrieval, will minimize uncertainties in the retrieval products. An ensemble of 216 cloud/atmosphere cases are used with an information content iterative routine and a clustering algorithm. Water vapor channels are identified as highly informative for cirrus cloud optical depth and cloud top pressure, away from window regions normally used in reflective solar ice cloud property retrievals. The ice-absorbing channels are most informative for effective particle size. The extractable independent information content increases with the increasing number of CLARREO-PF channels that are added to the retrieval system. However, with each channel added to the retrieval selection, the independent information content of all the remaining unselected channels decreases.**

***Index Terms*— Cirrus clouds, hyperspectral, remote sensing, retrieval.**

I. Introduction

CIRRUS clouds are upper tropospheric ice clouds that inhabit atmospheric layers with temperatures less than −30 °C. We know from work by Sassen et al. [1] that cirrus clouds have a global extent of 16.7% and a latitudinal frequency gradient, with frequencies of approximately 33% in the tropics down to 7% in polar regions. Given their heights in the atmosphere and global extent, they are important modulators of atmospheric radiant energy flows. Cirrus clouds can reflect incoming solar radiation back into space. This has the effect of increasing the Earth's albedo and cooling the Earth/atmosphere system beneath the cloud. They can also absorb longwave radiation emitted by lower atmospheric layers and the Earth's surface. Accurate knowledge of global cirrus radiative and microphysical quantities that represent how the clouds interact with radiation is best obtained through satellite retrievals. However, satellite ice cloud property retrieval products have large uncertainties that can impact downstream processes, such as energy budget calculations, climate modeling, and operational weather models [2], [3], [4]. High ice clouds are an important cause of intermodal variability [4], [5]. Waliser et al. [4] show that while other GCM variables, such as global mean precipitation and global mean precipitable water, have robust observational constraints, cloud ice constraints are still poor, a result confirmed by Eliasson et al. [3]. More robust satellite observations are required.

The purpose of this study is to objectively identify channel subsets in the spectral coverage of the Climate Absolute Radiance and Refractivity Observatory Pathfinder (CLARREO-PF) [6] instrument that when used as reflectance inputs to cirrus optical and microphysical property retrievals will minimize uncertainties in the retrieval products. CLARREO-PF is a hyperspectral instrument that measures ultraviolet, visible, shortwave infrared, and near-infrared reflectances. That is, CLARREO-PF produces spectrally resolved signals by measuring calibrated reflectances in many narrow and overlapping channels. Each measured pixel thus contains a highly information dense measurement, and utilizing the complete spectrum will maximize the information content for a retrieval. In addition, Cooper et al. [7] showed that fixing retrievals on just two or three channels, such as the Moderate Resolution Imaging Spectroradiometer (MODIS) dual-band retrieval [8], [9], is insufficient for all possible scenes encountered by ice cloud property retrievals. The authors showed that the subset of MODIS channels required to maximize the information for the retrieval, and thus reduce retrieval uncertainty, is highly dependent on the state of the atmosphere [7]. Thus, a hyperspectral instrument is better suited for maximizing the information given the possible

Manuscript received 10 July 2023; revised 21 December 2023 and 19 April 2024; accepted 13 May 2024. Date of publication 12 June 2024; date of current version 24 June 2024. This work was supported in part by the NASA Future Investigators in NASA Earth and Space Science and Technology Fellowship (FINESST) (Grant NNH19ZDA005K) and in part by the endowment funds associated with the David Bullock Harris Chair in Geosciences at Texas A&M University (account number 02-512231-00000). *(Corresponding author: Jeffrey C. Mast.)*

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Digital Object Identifier 10.1109/TGRS.2024.3413040

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cloud/atmosphere/surface combinations. The premise is that hyperspectral instruments will maximize the information content over that of multispectral instruments that spans a similar spectral region, given the hundreds to thousands of additional channels to choose from.

However, several downsides exist in using hyperspectral instruments for ice cloud property retrievals. First, utilizing the large number of channels in hyperspectral instruments creates large datasets, increasing the computational burden for retrievals. For example, lookup table-based retrievals will contain extremely large and unwieldy tables. In addition, optimal estimation retrievals require the forward model to calculate spectral reflectances and Jacobian matrices at each algorithm iteration creating a large computational burden. Second is the significant band-to-band redundancy of information. While King and Vaughan [10] showed that increasing the number of channels for a hyperspectral water cloud retrieval increases the total extractable information, they also showed that, with each added channel, the total independent information per additional channel decreases (see Fig. 6 in [11]). Therefore, it is suggested that 1) if a retrieval uses all the available channels from a hyperspectral instrument the information content will be maximized, and 2) the computational expense may not be acceptable (for an operational retrieval, for example) and it is better to identify channels that maximize the information content for the largest number of atmosphere/cloud/surface combinations while removing channels that minimize the information loss for the same combinations, that is, a tradeoff between retrieval accuracy and computational efficiency is necessary.

To find the optimal channel subset for ice cloud property retrievals, we use the information content of a measurement, also called the Shannon information content (hereafter SIC). Rodgers [11] in determining how to efficiently use the available radiance measurements in the AIRS instrument (an instrument with 2378 channels [12]) for atmospheric remote sensing posed the following questions: "Should all of the channels be used for operational retrievals or assimilations? Can a useful subset be chosen with minimum loss of information? What is the best way to select a subset of data?" Rodgers [11], applying the work of Shannon [13] to atmospheric retrieval characterization, describes the concept of the information content as a figure of merit that quantifies the amount "by which knowledge of a quantity is improved by making a measurement." The information content of a remote sensing instrument is calculated by subtracting the uncertainty of the retrieval system, characterized by its information entropy, from the uncertainty in the atmospheric state natural variability, also in the form of information entropy [11]. The information content can be understood as the ability of a retrieval system to reduce the probability density function representing the uncertainty of the retrieval variables. Thus, the channels that maximize the information content are those to use in a retrieval, while, based on project requirements, those that are discarded minimize the information loss. Furthermore, given that the information content is independent of the retrieval methodology and is a scalar figure of merit, it can be used to compare the relative measurement ability of instruments to determine the added value of a new instrument [11], which was later formally defined and derived as a means of retrieval diagnostics and optimization for atmospheric inversions [14].

The use of the SIC for optimal channel selection in satellite water and ice cloud remote sensing was later advanced by L'Ecuyer et al. [15], Cooper et al. [7], Wang et al. [16], and Chang et al. [17]. For example, Cooper et al. [7] used the Rodgers [11] methodology to objectively select MODIS shortwave and longwave channels for ice cloud optimal estimation retrievals. To this end, the authors created 800 synthetic atmospheric states spanning realistic ice clouds properties over ocean water. Their results showed that channel subsets in MODIS that maximized the SIC are atmospheric state-dependent [7]. Five channels were selected for retrievals, implemented, and verified in an optimal estimation retrieval framework [7], [18]. Results from the Cooper et al. [18] follow-on paper show that the retrieval derived from the SIC algorithm performed better than the MODIS bispectral retrievals. A separate MODIS SIC channel selection paper by Wang et al. [16] also suggested a five channel subset. However, the authors noted that more channels should be added based on the research requirements and tolerance for computational burden [16]. Chang et al. [17] expanded this methodology for the infrared hyperspectral AIRS instrument. The authors used a selection of 400 cloud/atmosphere cases to determine a channel subset that maximizes the SIC for the most cloud/atmosphere combinations while minimizing computational burden. Results showed that channel clusters around 14, 10.4, 4.2, and 3.8 μm contained the most information for the states investigated [19]. From this, the authors proposed a subset of 29 channels [19].

Current NASA hyperspectral instruments measuring UV, visible, shortwave infrared, and near-infrared reflected solar radiances include Airborne Visible/Infrared Imaging Spectrometer (AVIRIS, [20]), AVIRIS next-generation instrument (AVIRIS-NG), Earth Mineral Dust Source Investigation (EMIT, [21]), and CLARREO-PF. For this article, we limit our study to the CLARREO-PF, a forthcoming reflected solar hyperspectral spectrometer with a spectral coverage of 350–2300 nm, a 6 nm spectral resolution at 3 nm sampling [6]. Through a two-axis pointing system, CLARREO-PF will calibrate by viewing the sun and the moon, achieving an expected 0.3% (1-sigma) radiometric uncertainty [6]. While the mission goal of CLARREO-PF is to test and prove novel instrument technologies and methodologies for climate science, we believe that a cirrus optical depth, effective particle size, and cloud height retrieval will benefit from the upcoming highly accurate spectrally resolved measurements.

This article is organized as follows. The algorithm methodology used to select an optimal subset is described in Section II, including the information content algorithm, radiative transfer code, ice cloud scattering properties, and ancillary datasets. Note that the project is scoped for only single-layer ice clouds over ocean water with Lambertian scattering. Other surfaces, mixed clouds, and multilayer clouds will be investigated in future studies. Additionally, the channel selection is limited to the retrieval variables cloud optical depth (τ_{cloud}), effective particle diameter (D_{eff}), and cloud top pressure (P_{top}). As the reflectance sensitivities to retrieval variables and the retrieval uncertainties are important

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intermediate calculations for SIC, Section III, the results section, begins with these. A selection of 216 simulated cloud/atmosphere states over ocean water are used to select an optimal CLARREO-PF subset, of which an example is presented. Finally, Section III ends with a suggested channel subset for retrieval of cirrus cloud τ_{cloud}, D_{eff}, and P_{top} based on the frequency that channels are selected by the channel selection algorithm. A summary and conclusion of this article is given in Section IV, including a discussion of the results with respect to emerging fast radiative transfer models developed for hyperspectral calculations.

II. Methodology

The databases of the single-scattering properties used in the forward model, both as the assumed cirrus optical properties and in the calculations of uncertainties, are described in Section II-A. The forward model and needed ancillary datasets are given in Section II-B. Finally, the iterative algorithm for channel subsetting is described in Section II-C.

A. Ice Cloud Scattering Properties—Single-Scattering Property Models and Bulk Ice Cloud Scattering

The single-scattering properties of individual ice crystals, needed for the modeling of cirrus cloud optical and microphysical properties, are taken from two separate property databases. The Two Habit Model (THM, [22]) is the primary database in this study and the assumed optical properties of cirrus clouds. To estimate the uncertainties due to assumptions about the selection of a single-scattering property dataset, TAMUice2016 scattering library [23] is used. Both scattering libraries provide extinction efficiencies, single-scattering albedos, asymmetry parameters, and the scattering phase functions for modeled ice crystals at 396 discrete wavelengths from 0.2 to 15.25 μm for maximum diameters from 2.0 to 10 000 μm.

State-of-the-science THM database of the ice cloud single-scattering properties is in its third version [22]. The THM is based on a hypothesis that the optical properties of roughened hexagonal ice crystals can be best represented by the ensemble-averaged counterparts of randomly distorted ice crystals. The single-scattering properties calculated for the THM are ensemble means of a collection of 60 distorted hexagonal column particles and a collection of 20 particles made up of aggregates of 20 distorted hexagonal columns. To develop the THM single-scattering property database, a two stage calculation is performed using the Invariant Imbedding T-Matrix Method (II-TM; Johnson [24]; and Bi and Yang [23]) for small size parameters and the Improved Geometric Optics Model (IGOM; Yang and Liou [25]) for larger size parameters. Additionally, the Physical Geometric Optics Model (PGOM, Yang and Liou [26] and references cited therein) is used for the backscattering portion of the scattering phase function to produce more accurate backscattering calculations. The THM has been shown to achieve good spectral consistency in ice cloud retrievals and to have better backscatter results than existing models [22].

Bulk scattering properties for the cirrus clouds are calculated using the familiar definitions as in Baum et al. [27], Yang et al. [28], and Iwabuchi et al. [29]. For example,

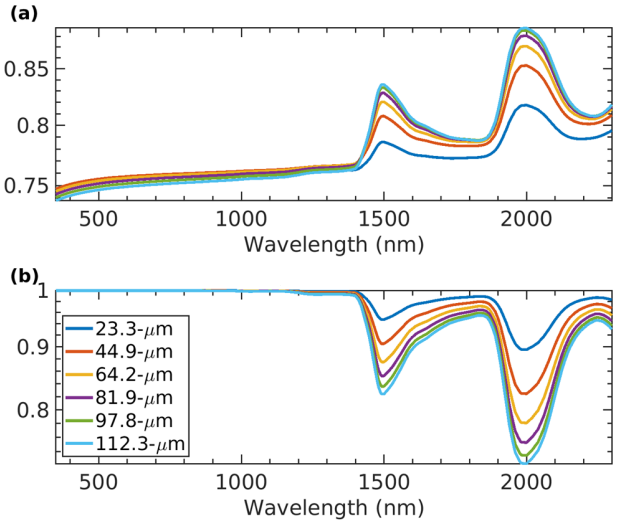


Fig. 1. Comparison of bulk scattering properties calculated for six cirrus clouds containing different effective diameter. These bulk properties are calculated using Two Habit Model single-scattering properties. Plot (a) is a bulk asymmetry factor and (b) is a bulk single-scattering albedo.

the unitless bulk single-scattering albedo $\langle\tilde{\omega}(\lambda)\rangle$, bulk asymmetry factor $\langle g(\lambda)\rangle$, and bulk scattering phase function $\langle p(\cos\Theta,\lambda)\rangle$ are

$$\langle\tilde{\omega}(\lambda)\rangle = \frac{\langle\sigma_{\text{scat}}(\lambda)\rangle}{\langle\sigma_{\text{ext}}(\lambda)\rangle} \tag{1}$$

$$\langle g(\lambda)\rangle = \frac{\int_{D_{\min}}^{D_{\max}} n(D)g(D,\lambda)\sigma_{\text{scat}}(\lambda)dD}{\int_{D_{\min}}^{D_{\max}} n(D)\sigma_{\text{scat}}(\lambda)dD} \tag{2}$$

$$\langle p(\cos\Theta,\lambda)\rangle = \frac{\int_{D_{\min}}^{D_{\max}} n(D)\sigma_{\text{scat}}(\lambda)p(\cos\Theta,\lambda)dD}{\int_{D_{\min}}^{D_{\max}} n(D)\sigma_{\text{scat}}(\lambda)dD}. \tag{3}$$

Here, $\sigma_{\text{scat}}(\lambda)$ is the scattering cross section (cm^2), $g(D,\lambda)$ is the unitless asymmetry parameter for particle maximum diameter D, and wavelength λ, $p(\cos\Theta,\lambda)$ is the unitless phase function for scattering angle Θ, and $\langle\sigma_x(\lambda)\rangle$ is the bulk interaction x (scattering or extinction) cross section (cm^2). The bulk optical properties are calculated with a gamma size distribution

$$n(D) = N_0\left(\frac{D}{\beta}\right)^{\alpha-1}e^{\frac{-D}{\beta}} \tag{4}$$

where α and β are specified as in [29].

The effective diameter is the representative crystal diameter based on the size distribution and is defined as

$$D_{\text{eff}} = \frac{3}{2}\frac{V_{\text{total}}}{A_{p,\text{total}}}. \tag{5}$$

The total volume V_{total} and total projected area $A_{p,\text{total}}$ are

$$V_{\text{total}} = \int_{D_{\min}}^{D_{\max}} V(D)n(D)dD \tag{6}$$

$$A_{p,\text{total}} = \int_{D_{\min}}^{D_{\max}} A_p(D)n(D)dD. \tag{7}$$

The bulk single-scattering albedo and asymmetry factor for the range of effective particle sizes in this study are shown in Fig. 1. In Fig. 2 are the scattering phase functions as a function of scattering angle [see Fig. 2(a)] and as a

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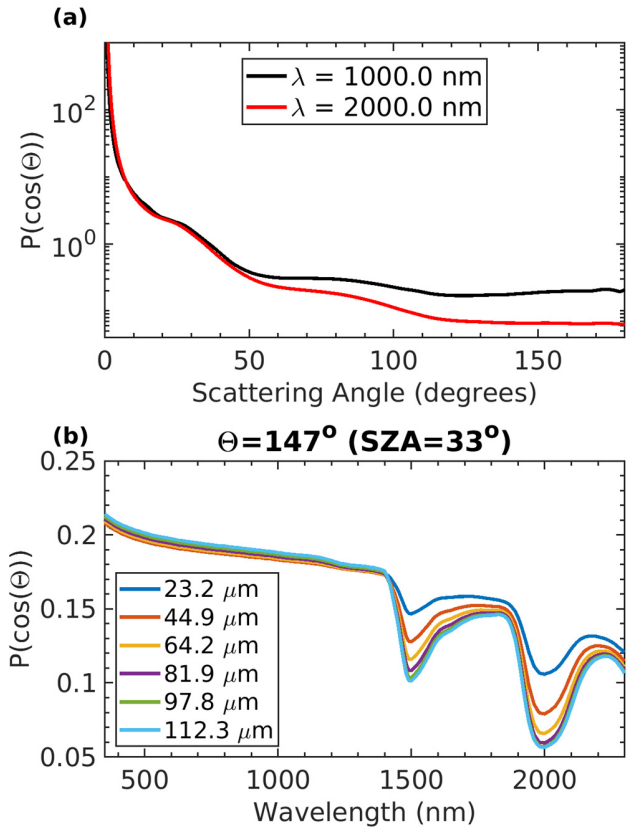


Fig. 2. Bulk scattering phase functions calculated for six cirrus clouds containing different effective diameters as a function of (a) scattering angle and (b) wavelength. These bulk scattering phase functions are calculated using Two Habit Model single-scattering properties. The scattering angle of 147° corresponds to the base case geometry of this study that includes a solar zenith angle of 33° and an instrument viewing at nadir.

function of wavelength for a scattering angle of 147° [see Fig. 2(b)]. The scattering angle in Fig. 2(b) corresponds to the sun-cloud-instrument base case geometry with SZA = 33° and the instrument viewing at nadir. Ice clouds scatter strongly and absorb weakly in the ultraviolet and visible, but become absorbing within the near-infrared. Major absorption features are present at approximately 1500 and 2000 μm corresponding to absorption by ice. For the case of the asymmetry factor, forward scattering increases slowly for all wavelengths, beginning at $\lambda = 350$ nm ($g \sim 0.74$ for all cases) through $\lambda \approx 1400$ nm, where the absorption bands for ice begin. The direction of the peaks is opposite in the cases of the asymmetry parameters and single-scattering albedos, indicating that both absorption and forward scattering increase together, leading to strong loss of photons (ocean water strongly absorbs forward scattered photons). Also, evident in these figures is that both are sensitive to the effective particle size in these bands.

B. Radiative Transfer Calculations and Postprocessing for Synthetic CLARREO-PF Radiances

A combination of two radiative transfer models, the Line-by-Line Radiative Transfer Model (LBLRTM; [30]) and the Discrete Ordinates Radiative Transfer (DISORT; [31]) model, hereafter LBLRTM+DISORT, is used to calculate top of atmosphere (TOA) reflected solar radiances in a two-stage approach. Such a model combination has been used in previous research as components of the forward model for retrievals of cloud phase and microphysical properties [32], [33], [34]. LBLRTM is used to calculate atmospheric gas optical depth. The inputs include vertical profiles of temperature, pressure, gas concentration, and molecular spectroscopic parameters from the AER v3.2-line parameter database. The AER line parameter database is based on the High-Resolution Transmission (HITRAN) database [35] with additional empirically determined adjustments [36]. The line parameter database contains information needed for construction of molecular absorption coefficients, including line strengths, line centers, and half-widths of atmospheric species. LBLRTM utilizes this information and the input atmospheric pressures and temperature to construct Voight line shapes at all atmospheric layers. The resultant absorption coefficients can be combined with the atmospheric gas concentrations to give the layer optical depths.

The vertical profiles of atmospheric temperature, pressure, and gas concentrations are obtained from the Modern-Era Retrospective Analysis for Research and Applications Version 2 (MERRA-2) reanalysis dataset [37]. MERRA-2 is a reanalysis dataset, or a merging of observations with modeling to create a blended historical meteorological dataset. MERRA-2 vertical profiles are used in this study as representative samples of atmospheric states. Profiles are collected from random locations in the North Atlantic during winter, Gulf of Mexico during summer months, and tropical Indian Ocean during one year, 2010. This was done to get a better representation of the possible atmospheres that a retrieval scenario will encounter. These profiles are further paired down to only those at local noon and zero cloud cover as reported from the cloud cover flag. The following variables are used from MERRA-2 (data collection in parentheses): CO and O_3 (inst3_3d_chm_Nv), temperature, pressure, surface pressure, relative humidity, and altitude (inst3_3d_asm_Nv), and surface temperature and geolocation data (inst1_2d_asm_Nx). The profiles are retained on the 72-level MERRA-2 pressure vertical grid, from 985.000 to 0.010 hPa [38]. The water vapor mixing ratio vertical profile needed for LBLRTM is then calculated from the relative humidity, temperature, and pressure. Four additional gases not in MERRA-2 (N_2O, CH_4, O_2, and CO_2) are obtained from a standard atmosphere matching the seasonal and location conditions of the acquired profile (tropical, midlatitude summer, midlatitude winter, etc.) [39].

DISORT is run from 4344.0 to 28819.0 cm^{-1} at 1 cm^{-1} wavenumber steps. DISORT models the scattering phase function through a Legendre polynomial expansion with the user supplying the coefficients of the Legendre polynomial expansion. In this study, an $N = 6000$ term expansion is used. DISORT utilizes the discrete ordinates numerical method for solving the radiative transfer equation by reducing the infinite number of possible directions a beam can take, and subdivides 4π steradians into a finite, user-defined number of beam directions in a slab geometry. DISORT v4.4 is used in this article with 32 beam directions.

To simplify radiation interactions with the surface, all calculations are performed over ocean surfaces with Lambertian reflectance. The USGS Spectral Library Version 7 [40] open

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ocean water is used for surface spectral albedo. The Naval Research Lab Solar Spectral Irradiance version 2 (NRLSSI2) [41] provides TOA downwelling spectral flux specific to the dates of the chosen MERRA-2 profiles.

C. Ice Cloud Information Content Retrieval Optimization

In seeking the optimal channels for an ice cloud retrieval for the CLARREO-PF instrument, we leverage the information content algorithm of Rodgers [11]. Here, the "Shannon information content," or "measurement information content," is a figure of merit for knowledge gained or, conversely, uncertainty reduced [11]. In the equations below, capital bold letters represent matrices, lowercase bold represent vectors, and nonbold letters represent scalar variables. The scalar SIC is represented by the difference of Shannon entropies (H) of two atmospheric states

$$\mathrm{SIC} = H_1 - H_2 \tag{8}$$

where H_1 is entropy of the state before a measurement, and H_2 is the entropy of the state given the measurement. From [14], the Shannon entropy can be thought of as a quantity of uncertainty, and is

$$H[P] = -\sum_i P_i \log_2(P_i) \tag{9}$$

where P_i is the probability of state "i." Rodgers [14] further showed that for an observing system measuring reflectance in a series of channels, represented here by vector $\boldsymbol{y}$, and a retrieval state described by a vector $\boldsymbol{x}$ (here $\{\tau_{\mathrm{cloud}}, D_{\mathrm{eff}}, P_{\mathrm{top}}\}$) the SIC can be calculated completely from measurement sensitivities, measurement and forward model uncertainties, and prior state uncertainties

$$\begin{aligned}\mathrm{SIC} &= H[P(\boldsymbol{x})] - H\big[P(\boldsymbol{x}|\boldsymbol{y})\big] \\ &= \frac{1}{2}\log_2\left|\left(\boldsymbol{K}^T \boldsymbol{S}_\epsilon^{-1}\boldsymbol{K} + \boldsymbol{S}_a^{-1}\right)\boldsymbol{S}_a\right|.\end{aligned} \tag{10}$$

In the above, $\boldsymbol{S}_a$ and $\boldsymbol{S}_\epsilon$ are the prior and error covariance matrices, respectively, and $\boldsymbol{K}$ is the forward model Jacobian.

The prior covariance matrix, $\boldsymbol{S}_a$, is a 3×3 matrix that is constructed using climatological data representing the natural variability of the state. This represents what we know of the scene before a measurement. Here, one year of the MODIS Collection 6 ice cloud products for τ_{cloud}, D_{eff}, and P_{top} are used to populate the matrix. The MODIS sensors aboard the Terra and Aqua platforms have been collecting data since December 19, 1999, and May 4, 2002, respectively [8]. The MODIS science team estimates τ_{cloud} and D_{eff} using a dual channel reflected solar retrieval [9] whereby one channel measures radiance in a nonabsorbing region (0.66, 0.86, or 1.24 μm) that is sensitive to τ_{cloud}, and an absorbing channel (1.6, 2.1, or 3.7 μm) that is sensitive to D_{eff}. The results are determined by a comparison of measured reflectances to precomputed lookup tables mapping reflectance to τ_{cloud} and D_{eff}. Inferring cloud top pressure is based on a combination of the 15 μm band CO_2 slicing method [42] and the National Centers for Environmental Prediction's Global Data Assimilation System [43], [44].

$\boldsymbol{S}_\epsilon$ is the sum of the measurement and modeling error covariance matrices

$$\boldsymbol{S}_\epsilon = \boldsymbol{S}_I + \boldsymbol{S}_{\mathrm{WV}} + \boldsymbol{S}_T + \boldsymbol{S}_{\mathrm{Habit}} \tag{11}$$

where $\boldsymbol{S}_I$ is the instrument error covariance matrix, $\boldsymbol{S}_{\mathrm{WV}}$ is the error covariance matrix of the forward model results due to the uncertainties in the water vapor input, $\boldsymbol{S}_T$ is the error covariance matrix of the forward model results due to the uncertainties in the temperature input, and $\boldsymbol{S}_{\mathrm{Habit}}$ is the error covariance matrix due to habit uncertainty.

The Jacobian matrix, $\boldsymbol{K}$, contains 640 rows, one for each CLARREO-PF channel, and three columns, one for each retrieval variable. To populate the Jacobian, a series of sensitivity studies are conducted for each of the variables the channel selection is being optimized for. The parameters τ_{cloud} and D_{eff} are each perturbed by 0.25% while P_{Top} is perturbed by one MERRA-2 pressure layer. When one of the parameters is perturbed, all others remain fixed. For example, when the radiative transfer model is run for channel 10 and the sensitivity of the reflectance for cloud optical depth is tested, then the Jacobian component for channel 10 and parameter τ_{cloud} is $\boldsymbol{K}_{10,\tau_{\mathrm{cloud}}} = (\Delta R_{10,\tau_{\mathrm{cloud}}}/\Delta\tau_{\mathrm{cloud}})$. Here, $\Delta R_{10,\tau_{\mathrm{cloud}}}$ is the change in reflectance in channel 10 due to perturbing τ_{cloud} by 0.25% and $\Delta\tau_{\mathrm{cloud}}$ is the change in optical depth.

By convention, the information content is calculated with log base-two and the units are bits. Thus, the number of possible states in the result is 2^{SIC}, and one bit of information corresponds to reducing the volume of the probability distribution representing the uncertainties in the retrieval variables by 1/2. The larger the SIC, the more information a measurement provides, and the better a measurement is able to resolve the atmospheric state being observed. L'Ecuyer et al. [15] likened this to applying tick marks to a ruler measuring one of the variables of interest. The larger the extracted information content, the finer one can mark ticks in a ruler.

To isolate the most informative channels, it is assumed that retrieval channel errors are uncorrelated. Then, from Chang et al. [17], (10) can be written as

$$\mathrm{SIC}_i = \frac{1}{2}\log_2\left|\boldsymbol{S}_a\left(\boldsymbol{k}_i^T \sigma_i^{-2}\boldsymbol{k}_i + \boldsymbol{S}_a^{-1}\right)\right|. \tag{12}$$

Here, subscript "i" signifies channel i of the CLARREO-PF instrument. SIC_i is the measurement information content in channel i, and we use $\boldsymbol{k}_i$ to represent ith row vector from the Jacobian $\boldsymbol{K}$. That is, $\boldsymbol{k}_i$ is the sensitivity of channel i to the retrieval variables τ_{cloud}, D_{eff}, and P_{Top}. The scalar value σ_i is the error standard deviation for channel "i." This is taken from element $\boldsymbol{S}_\epsilon(i,i)$ of the error covariance matrix $\boldsymbol{S}_\epsilon$. The parameter contains results from uncertainty studies for each channel, both from the forward model and the instrument. From (11)

$$\sigma_i = \sqrt{\delta R_{I,i}{}^2 + \delta R_{\mathrm{WV},i}{}^2 + \delta R_{T,i}{}^2 + \delta R_{\mathrm{Habit},i}{}^2}. \tag{13}$$

Once the sensitivities and uncertainties are established, an information spectrum is constructed using (12). From this information spectrum, the channel with the greatest SIC is identified. To identify the channel with the second most independent SIC, the prior state is constrained based on the posterior covariance matrix of the first channel

$$\boldsymbol{S}_{a,\mathrm{new}} = \left(\boldsymbol{k}_i^T \sigma^{-2}\boldsymbol{k}_i + \boldsymbol{S}_a^{-1}\right)^{-1}. \tag{14}$$

The SIC spectrum is recalculated and the next channel with maximum SIC can be selected from the remaining channels.

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The first selection will have the greatest information, the second less than the first, and so on. This continues until all channels are exhausted or a stopping condition is met.

To identify a stopping threshold for the iteration, the diagonal components of the "averaging kernel matrix" will be used. Rodgers [14] defines the averaging kernel matrix as the derivative of the retrieval state to the true state of the atmosphere

$$\boldsymbol{A} = \frac{\partial \hat{\boldsymbol{x}}}{\partial \boldsymbol{x}} = \left(\boldsymbol{K}^T \boldsymbol{S}_\epsilon^{-1} \boldsymbol{K} + \boldsymbol{S}_a^{-1}\right)^{-1} \boldsymbol{K}^T \boldsymbol{S}_\epsilon^{-1} \boldsymbol{K} \tag{15}$$

where $\boldsymbol{K}$ is the previously discussed forward model Jacobian, and $\hat{\boldsymbol{x}}$ denotes the retrieval of an atmospheric quantity with $\boldsymbol{x}$ the true value. For an ideal retrieval, $\boldsymbol{A} = \boldsymbol{I}$, the identity matrix.

Rodgers [14] interprets the diagonal of $\boldsymbol{A}$ as a weighting of the information, though the diagonal of $\boldsymbol{A}$ can also be interpreted as a sensitivity of the retrieved state to the true state

$$\boldsymbol{A}(1,1) = \frac{\partial \hat{\tau}_{\text{Cloud}}}{\partial \tau_{\text{Cloud}}} \tag{16}$$

$$\boldsymbol{A}(2,2) = \frac{\partial \hat{D}_{\text{eff}}}{\partial D_{\text{eff}}} \tag{17}$$

$$\boldsymbol{A}(3,3) = \frac{\partial \hat{P}_{\text{top}}}{\partial P_{\text{top}}}. \tag{18}$$

If $\boldsymbol{A}(1,1) = \partial \hat{\tau}_{\text{Cloud}}/\partial \tau_{\text{Cloud}} = 1$, then the retrieval of cloud optical depth is highly sensitive to the true cloud optical depth. However, if $\boldsymbol{A}(1,1) = \partial \hat{\tau}_{\text{Cloud}}/\partial \tau_{\text{Cloud}} = 0$, then there is no information to extract on cloud optical depth. Rodgers [14] further defines the sum of the diagonal of $\boldsymbol{A}$ as the "degrees of freedom for signal" (df_s), which for an ideal retrieval is rank($\boldsymbol{K}$). For an actual retrieval, $df_s <$ rank($\boldsymbol{K}$). In this way, iteration can be terminated once the "sensitivity" of each of the retrieval variables to the true state, or all the degrees of freedom for signal components, crosses below a desired threshold. In this article, the threshold of termination is chosen to be $\boldsymbol{A}(i,i) < 0.5$. That is, when $\boldsymbol{A}(1,1) < 0.5$ AND $\boldsymbol{A}(2,2) < 0.5$ AND $\boldsymbol{A}(3,3) < 0.5$, the iteration process will halt. A flowchart of the selection algorithm is presented in Fig. 3.

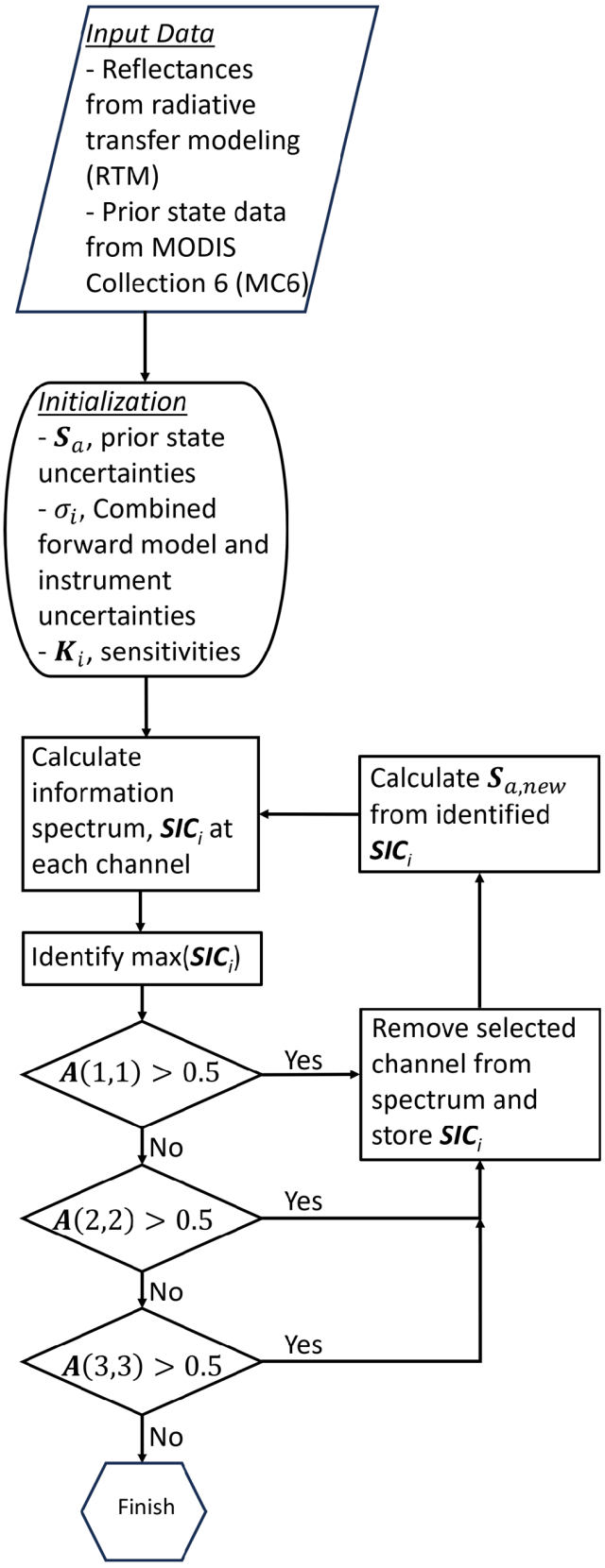


Fig. 3. Flowchart of the channel subsetting algorithm used in this article.

III. Results

In this section, we present Shannon information content results and the required supporting sensitivity and combined uncertainty calculations. A selection of 216 simulated cloud/atmosphere states over ocean water are used to select an optimal CLARREO-PF subset. These states included different atmospheric conditions, cloud optical depths, effective particle sizes, and scattering geometries (though the instrument is always viewing nadir). It is impractical to present results for all 216 cases run in this study, so an example case using data from an August 30, 2010, MERRA-2 profile collected over the Gulf of Mexico is shown. For the base case, a series of optically thin ice clouds ($\hat{\tau}_{\text{Cloud}} = \{0.1, 0.4, 0.6, 0.8, 1.0, 2.0\}$, $D_{\text{eff}} = \{23.3, 44.9, 64.2, 81.9, 97.8, 112.3\ \mu\text{m}\}$) are modeled with a top height at approximately 12 km ($P_{\text{top}} = 244.9$ hPa) over an ocean surface with Lambertian scattering. The solar zenith angle is set to 33° and the instrument is viewing nadir. First, in Section III-A, representative results of the sensitivity studies, used to populate the $\boldsymbol{K}$ Jacobian matrices, are presented. This is followed by a representative uncertainty study used to populate the combined uncertainty variable, σ_i,

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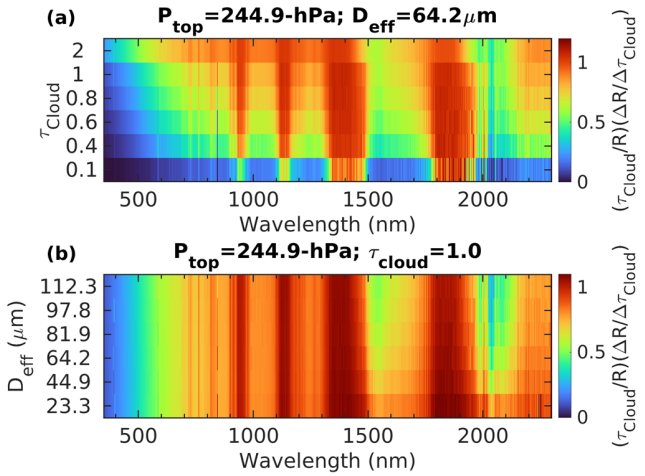


Fig. 4. Sensitivity of reflectance to perturbations of ice cloud optical depth. Plot (a) is a sensitivity spectrum as a function of wavelength and cloud optical depth. The y-axis denotes the optical depth for six cirrus clouds that have fixed $D_{\text{eff}} = 64.2$ μm. Plot (b) is also a sensitivity spectrum with the y-axis denoting D_{eff} for fixed $\tau_{\text{cloud}} = 1.0$. The ice cloud was placed at approximately 12 km ($P_{\text{top}} = 244.9$ hPa) in an August 30, 2010, Gulf of Mexico atmospheric profile from MERRA-2.

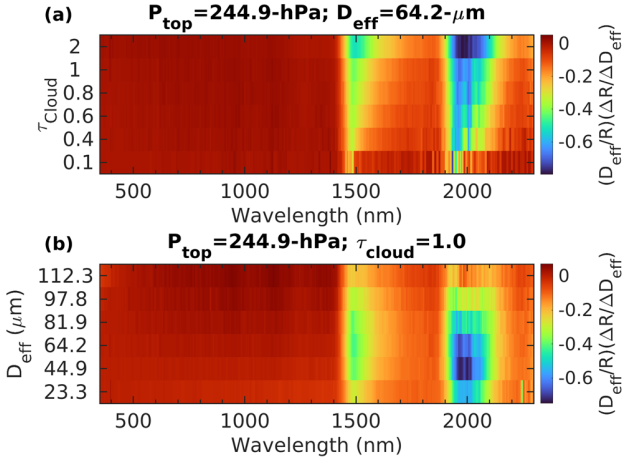


Fig. 5. Plots here are like those in Fig. 4, except the sensitivity in TOA reflectance here is due to perturbation of the cirrus cloud effective diameter. Plot (a) sensitivity spectrum with the y-axis representing the optical depths for six cirrus clouds that each have a fixed effective diameter of 64.2 μm. Plot (b) is the sensitivity spectrum with the y-axis representing the effective diameters of six cirrus clouds each with a fixed optical depth of 1.0.

in Section III-B. The resulting information content analysis is shown in Section III-C, and a subset of channels for a cirrus cloud retrieval is put forth in Section III-D.

A. Sensitivity Study of TOA Reflectance to Retrieval Variables

Fig. 4 has two contour plots showing the sensitivity of reflectance calculations to perturbations in cloud optical depth. All clouds are cirrus over ocean water (a dark, highly absorbing surface). Fig. 4(a) is a sensitivity spectrum as a function of wavelength and optical depth. Wavelength is along the x-axis. The y-axis can be thought of as six separate clouds. The effective diameter for each cloud is fixed at 64.2 μm, and the cloud top pressure is fixed at 244.9 hPa ($\sim$12 km). For each of the synthetic cirrus clouds, τ_{cloud} is perturbed by 0.25%. The sensitivities have been normalized such that the z scale color bar is $(\tau_{\text{cloud}}/R)(\Delta R/\Delta\tau_{\text{cloud}})$. Starting at the bottom with the extremely optically thin cloud, $\tau_{\text{cloud}} = 0.1$, sensitivity in reflectance at TOA is lower than for the optically thicker clouds in most bands for the perturbation of τ_{cloud}. More photons penetrate this cloud and are absorbed by lower water vapor or the surface. As the base optical depth increases, so does the sensitivity to the perturbation. From left to right, the sensitivity increases with increasing optical depth in the visible where single-scattering albedo is 1. Of note are the water vapor bands centered at 0.935, 1.13, 1.38, and 1.88 μm that show strong sensitivity to a perturbation of cloud optical depth for all y-axis cases. Water vapor is highly absorbing and predominately lies below high cirrus clouds such as these synthetic clouds at $\sim$12 km. For this reason, increasing cloud optical depth decreases absorption below the clouds within these channels. Also of note is the presence of sensitivity in the oxygen A-band at $\sim$760 nm. This band is a source of research in cloud top height [45], [46].

Similarly, Fig. 4(b) has six clouds represented on the y-axis, but this time each cloud has a different effective particle size. The optical depth for the clouds in this lower plot is fixed at $\tau_{\text{cloud}} = 1.0$. As in the upper plot, τ_{cloud} is perturbed by 0.25%, and the effect of this as a function of D_{eff} on the reflectance is noted. Again, the water vapor bands show large sensitivity. Note in the UV, visible, and most of the infrared, the sensitivity is stable among clouds across the different effective diameters. That is, for a fixed optical depth, the same perturbation of τ_{cloud} leads to a relatively similar relative sensitivity for all the clouds as a function of D_{eff}. Two areas where this is not the case are the locations of the ice absorption centered around $\sim$1500 and 2000 nm. With increasing diameter, the sensitivity, while positive, is decreasing relative to the surrounding channels. With increasing effective diameter, the bulk cloud becomes more absorptive in these ice absorption bands. Additionally, photons are increasingly scattered in the forward direction to the highly absorptive ocean water. This agrees with the figure of bulk cloud properties (see Figs. 1 and 2) which shows troughs for the single-scattering albedo (increased absorption) and peaks for the asymmetry factor (increased forward scattering).

Sensitivities of the TOA reflectances to perturbations of cloud D_{eff} are presented in Fig. 5. These plots are like those in Fig. 4, with wavelength along the x-axis and the y-axis representing six separate clouds. The exception is that all parameters are held fixed except D_{eff}, which is perturbed by 0.25%. In the Fig. 5(a), which contains τ_{cloud} along the y-axis, sensitivity is near zero in the UV, visible, and much of the near-infrared. Ice clouds are nonabsorbing in these bands; thus, there is minimal response here to the perturbation of D_{eff} (not exactly zero, but slightly negative as there is a slight increase in forward scattering). The exception is in the ice absorption bands. In the absorption bands the sensitivity is negative, signifying loss of photons. From Figs. 1 and 2, perturbing the ice cloud D_{eff} leads to an increase in absorption and forward scattering of the photons toward the surface, a dark ocean surface in this case. Fig. 5(b) is similar, but for fixed optical depth and clouds with varying D_{eff} (sizes are prior to perturbation). The major features in these figures again correspond to the locations of troughs in the bulk scattering properties with the resultant sensitivity being negative.

The final retrieval variable is the cloud top pressure. The sensitivity of the reflectance to perturbation of this parameter is shown in Fig. 6. Here, the cloud has been raised up in altitude by one MERRA-2 layer, corresponding to about a

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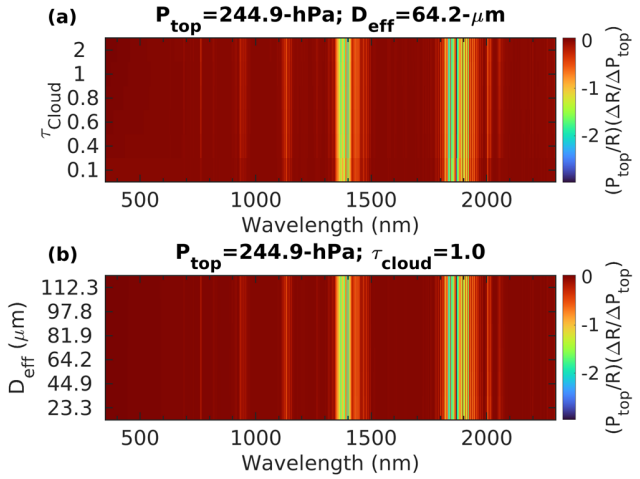


Fig. 6. Plots here are like those in Figs. 4 and 5, except the sensitivity in TOA reflectance here is due to perturbation of the cirrus cloud top pressure. Plot (a) sensitivity spectrum with the y-axis representing the optical depth for six cirrus clouds that each have a fixed effective diameter of 64.2 μm. Plot (b) is the sensitivity spectrum with the y-axis representing the effective diameters of six cirrus clouds each with a fixed optical depth of 1.0.

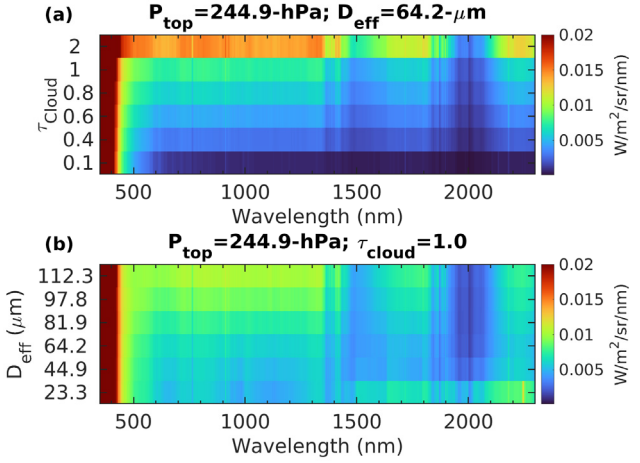


Fig. 7. Combined uncertainty spectra. (a) An uncertainty spectrum as a function of wavelength and cloud optical depth. The y-axis denotes the optical depth for six cirrus clouds that have fixed $D_{\text{eff}} = 64.2$ μm. (b) An uncertainty spectrum with the y-axis denoting D_{eff} for fixed $\tau_{\text{cloud}} = 1.0$. The atmospheric scene is the same as in Figs. 4–6.

1 km increase in cloud top height. This decreases the cloud top pressure from 244.2 to 208.2 hPa and the resultant sensitivity is $(P_{\text{top}}/R)(\Delta R/\Delta P_{\text{top}})$. The effect of raising the cloud in height is to decrease the amount of absorbing gas above the cloud, the major ones being water vapor and oxygen. This can be seen in the figure as the water vapor, and oxygen absorption bands show increasing negative sensitivities due to the reduction in cloud top pressure, thus negative ΔP_{top}, in the denominator.

B. Uncertainty Study

The final component required for the computation of the information content of the CLARREO-PF instrument is an analysis of the instrument and forward model uncertainties. The instrument uncertainties, $\delta {R_I}^2$, are from a field campaign of the HyperSpectral Imager for Climate Science (HySICS) instrument, a precursor for the CLARREO-PF instrument [47]. These are used as a proxy for CLARREO-PF uncertainties.

The modeling uncertainties consist of the uncertainty in choosing a scattering property database and uncertainties in the other "ancillary" inputs. Here, the ancillary inputs are chosen to be $\delta {R_{\text{WV}}}^2 + \delta R_T^2$, the uncertainties in the TOA reflectances due to the uncertainties in the water vapor and temperature. The water vapor uncertainty is calculated from relative humidity and temperature profiles, both obtained from MERRA-2. The uncertainties in the published MERRA-2 relative humidity (10% from the surface to 500 hPa, and 30% from there to TOA) and temperature (2 K) are taken from Luo et al. [48]. The uncertainties in these profiles are propagated from calculation of water vapor mixing ratio for input into LBLRTM to its effect on TOA radiances from DISORT following the Law of Propagation of Uncertainty as in Taylor and Kuyatt [49].

Determining the reflectance uncertainty inherent in choosing a scattering model is more difficult given that there is no accepted manner of calculating this. In this study, the "habit uncertainty" is estimated as the variance of five separate synthetic TOA reflectances, each calculated using a different ice habit model. The THM is considered the base case for the study, with the remaining four habits coming from TAMUIce2016. TAMUIce2016 contains the single-scattering properties for 11 ice crystal habits: hexagonal columns, hollow hexagonal columns, plates, droxtals, aggregates of five plates, aggregates of ten plates, aggregates of eight hexagonal columns, solid bullet rosettes, hollow bullet rosettes, spheres, and spheroids [23]. Additionally, TAMUIce2016 accounts for three levels of possible crystal surface roughness. The included roughness factor has the effect of smoothing the phase function, enhancing the retrieved D_{eff}, and decreasing the retrieved τ_{cloud} [50]. Thus, with TAMUice2016, we can model additional cirrus cloud optical and microphysical properties using common habits in published literature. In this way, we can account for the uncertainty due to assumptions made when researchers select a habit for radiative transfer calculations. The following habit shapes from TAMUIce2016 are selected for the uncertainty calculation: smooth hexagonal column habits, hexagonal column habits with severely roughened surfaces, aggregates of hexagonal columns with smooth surfaces, and aggregates of hexagonal columns with severely roughened surfaces.

The combined uncertainty is calculated as (13). This final product can be seen in Fig. 7. As in the sensitivity calculation figures, Fig. 7(a) was calculated with a fixed D_{eff} with six separate cloud optical depths along the y-axis, while Fig. 7(b) has a fixed τ_{cloud} with six D_{eff}'s along the y-axis. One important feature is in the UV and visible where the instrument uncertainty is large (see published HySICS uncertainties in [47]). To enhance other features, the uncertainty scale in the z-axis is cut off at $12 \times 10^{-3} W \times \text{m}^{-2} \times \text{SR} \times \text{nm}^{-1}$. This causes it to show as "off the scales." The remaining features are due to the habit uncertainty, as outside of the ultraviolet, it is the largest uncertainty within the reflected solar spectra. In the upper panel, as optical depth increases, the combined uncertainty increases throughout most of the visible and near-infrared. Additionally, in both Fig. 7(a) and (b) have minima in the absorption bands.

C. Information Content of CLARREO-PF Spectra

Separate information content analyses were conducted for 216 cirrus cloud cases. Again, a representative example

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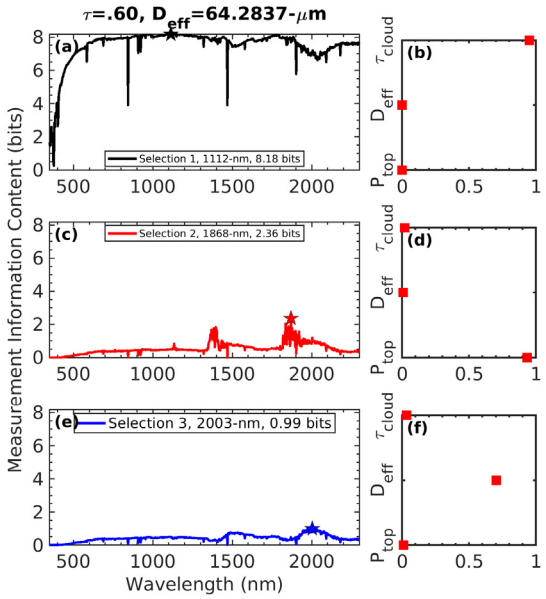


Fig. 8. Information spectra calculated for a radiative transfer case with a MERRA-2 atmosphere over the Gulf of Mexico on 8/30/2010, and a cloud with $\tau_{\text{cloud}} = 0.60$, $D_{\text{eff}} = 64.28$ μm, $P_{\text{top}} = 244.9$ kPa. The plots in the left column, (a), (c), and (e) are information spectra. The stars denote the channel locations with the maximum independent information. The plots in the right column, (b), (d), and (f), contain the corresponding *dfs* for the SIC spectrum directly to its left.

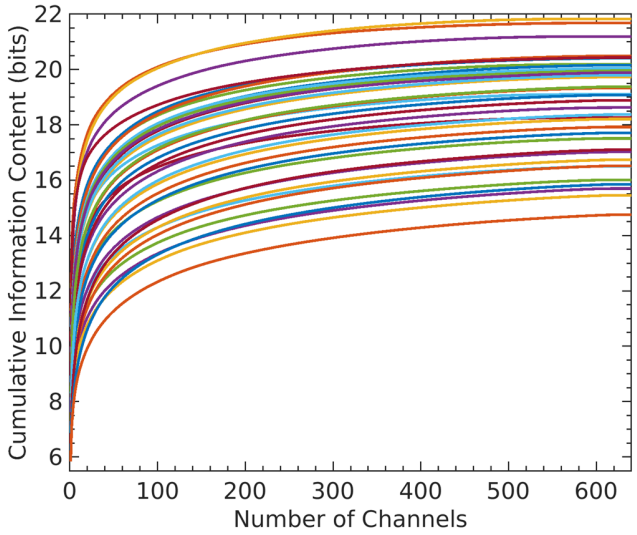


Fig. 9. Cumulative independent extractable information for 36 separate cloud cases. This was calculated for all 640 channels using the iterative method. The atmosphere was held constant with the MERRA-2 summer (8/30/2010) Gulf of Mexico atmosphere that was used for Figs. 8–11.

will be presented. In Fig. 8 are six plots corresponding to three channel selections using the iterative method described in Section II-C for a single cloud case. The plots in the left column pertain to information content spectra or the information content per wavelength, for synthetic CLARREO-PF reflectances. These information content spectra were calculated for a scene with a model cirrus cloud placed with its top at approximately 12 km ($P_{\text{Top}} = 244.9$ hPa), $\tau_{\text{cloud}} = 0.6$, and effective diameter $D_{\text{eff}} = 64.28$ μm. The atmosphere is an 8/30/2010 Gulf of Mexico open water profile from MERRA-2. The sun is located at 33° from the zenith and the instrument is viewing at nadir. The x-axis is the wavelength of the spectrum, and the y-axis is the information content with units of bits. Fig. 8(a) is the information spectrum available at iteration step 1. Several important features present in this figure will now be pointed out. First, at the ultraviolet end are channels with minimal extractable information due to the instrument uncertainty being very large there. Next, the maximum information content location in the spectrum has been marked with a star at 1112 nm, with 8.18 bits of extractable information. The information most likely corresponds to retrieval variable τ_{cloud} (see Fig. 3). This can be confirmed in Fig. 8(b) where a plot of the degrees of freedom for signal components for the channel is marked by the star in Fig. 8(a). This shows that $\boldsymbol{A}(1,1) = 0.99$, $\boldsymbol{A}(2,2) = 0.0$, and $\boldsymbol{A}(3,3) = 0.0$; so a retrieval utilizing this channel will strongly characterize the optical depth as a retrieval target.

From Fig. 4, the CLARREO-PF water vapor channels throughout the reflected solar spectrum are highly sensitive to τ_{cloud}. This means the information in the channels can be strongly correlated. The selection of one channel will limit the independent information in other surrounding channels. To extract additional independent information (14) is used to constrain the prior state, and following this, the information spectrum is recalculated. The resultant SIC spectrum has only information independent of the prior selected channel and is shown in Fig. 8(c). A red star marks the maximum information at the 1868 nm channel with 2.36 bits. As shown in the Fig. 8(d), this information is tied to P_{top}. From Fig. 6, there are several channels within two "bands" in the spectrum that are highly sensitive to cloud top pressure. These bands are correlated, and constraining the prior uncertainty will again remove redundant information. Fig. 8(e) contains information on the effective particle diameter at 2003 nm for 0.99 bits. Notice that the number of degrees of freedom for signal for this selection is greater than 0.5, so iteration will continue until a channel is selected whose *dfs* components fall below the threshold.

D. Channel Selection Recommendations for CLARREO-PF Ice Cloud Retrievals

In all the above examples, notice that at each iteration step (selection 1 → selection 2 → selection 3), the magnitude of the extractable independent information decreases with each selection. This can be seen in Fig. 9, a plot of the cumulative independent information for 36 separate cloud state cases, that is, each case has a different τ_{Cloud}, D_{eff}, and/or P_{top}. The cumulative information is calculated for all 640 CLARREO-PF channels. In all cases, the 08/30/2010 Gulf of Mexico atmosphere is used.

Fig. 9 shows a problem of diminishing returns in independent information content with the addition of channels to a retrieval. This problem was also noted in King and Vaughan [10] for water clouds. The first channel selected has the maximum information content. However, as shown in Fig. 9, with each channel selected, the remaining channels have less information available. This reduced independent information with each selection eventually becomes asymptotic as the remaining channels are in spectral locations with high noise (e.g., the UV, where instrument noise is high, see Fig. 7) or very low sensitivity.

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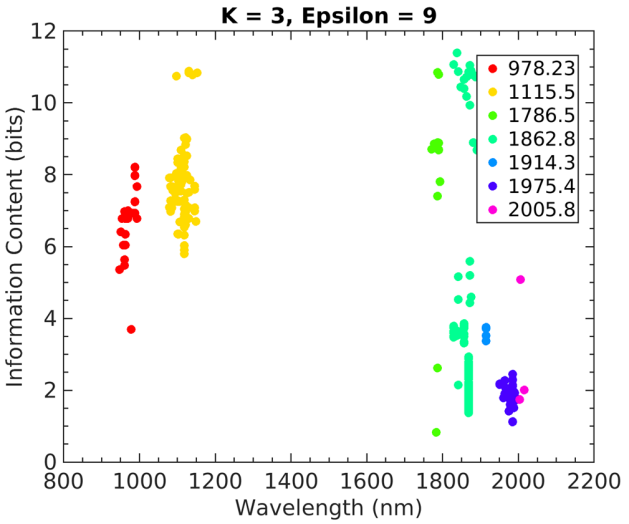


Fig. 10. Application of DBSCAN clustering algorithm to the channels selected by the iterative algorithm.

While we recommend using as many channels as is computationally efficient, when using rigorous physical models, as in this article, this creates a difficult problem due to the computational burden. The goal of this article is to find the subset of channels for most combinations of cloud/atmosphere states that will alleviate this computational burden. A total of 216 cloud/atmosphere cases were modeled and the iterative algorithm of Section II-C found the channels with high information content in each individual cloud/atmosphere case. Following this, a clustering algorithm is used to find channel clusters in SIC versus channel wavelength space.

The density-based spatial clustering of applications with noise (DBSCAN, [51]) algorithm is used to identify channel clusters based on their frequency of selection across the different cloud/atmosphere cases. DBSCAN is a form of unsupervised statistical classification that identifies clusters in data based on the straight-line distance between neighboring points. The user specifies the minimum number of data points within a cluster (k) and a minimum radius of the clusters (ε). According to Schubert et al. [52] DBSCAN operates as follows: A "core point" is any data point in {wavelength, SIC} space that is connected to, at minimum, k other points within a radius ε. Any point connected to a core point that has less than k points connected to it is considered a border point. Any point not connected to either core points or border points of the cluster is considered noise and discarded. From [51], [52] $k = 4$, or twice the number of dimensions. The heuristic for selection of ε is based on the methodology used in Rahmah and Sitanggang [53], where the mean distance between each point and its k-nearest neighbors is calculated and sorted. In a line drown in {channel count, mean k-nearest neighbor distance} space, the optimal selection for ε is the point of maximum curvature.

The result of using DBSCAN in {wavelength, SIC} space is shown in Fig. 10, producing one possible subset of channels for a retrieval over ocean water. The legend in the top right of the plot notes the wavelength of the centroids of the clusters. Most of the clusters are situated around strong water vapor bands including those with centroids at 978.23, 1115.5, 1786.5, 1862.8, and 1914.3 nm. Additionally, the 1975.4 and 2005.8 nm clusters contain D_{eff} sensitive areas in highly absorbing ice cloud channels.

IV. Summary and Conclusion

Considering new and anticipated hyperspectral reflective solar satellite instruments, an analysis of the information content of the CLARREO-PF for cirrus cloud optical and microphysical property retrievals was conducted. The purpose of this study is to objectively select an optimal subset of the complete CLARREO-PF channel range that will maximize the information content of the instrument for a retrieval of τ_{Cloud}, D_{eff}, and P_{Top}. The reason to subset CLARREO-PF channels for a cirrus cloud retrieval is due to the computational burden imposed by the forward model simulations of the hyperspectral radiances. As the information content can be completely described by the measurement's sensitivity and retrieval system uncertainty, as well as the natural variability of the target variables, in depth sensitivity and uncertainty studies are presented. To determine the optimal channel subset for a cirrus optical and microphysical property retrieval, the iterative method suggested by Rodgers [11] and specifically based on the work of Cooper et al. [7], L'Ecuyer et al. [15], and Chang et al. [17], is used. Iteration is halted when $dfs \leq 0.5$ for all retrieval variables. Using the DBSCAN clustering algorithm, clusters of channels are identified representing channels that the iterative algorithm selected at high frequency. We suggest, for CLARREO-PF ice cloud retrievals, that scientists use as many channels as computational expense is acceptable. However, we put forth a selection of seven channels based on the cluster centroid channels. It is our belief that more work is needed, especially modeling atmospheric states over land surfaces and nonidealized ocean water. Additionally, this study scoped the cirrus clouds to $\tau \leq 3.0$. Future work will expand the cloud optical depth to $\tau = 10.0$ matching the International Satellite Cloud Climatology Project (ISCCP) definition of cirrus clouds [54].

Three main conclusions can be drawn from this study. First, the water vapor bands are highly informative for cirrus cloud optical depth and, to a lesser extent, cloud top pressure. Using a selection of 216 synthetic cloudy scenes, the channel selection algorithm returned channel clusters centered at 0.978, 1.115, 1.786, and 1.862 μm as having high information content. The sensitivity and information content in the water vapor channels is understandable given that cirrus clouds are upper tropospheric clouds. This height puts them above the majority of atmospheric water vapor. As water vapor bands in the near-infrared are highly absorbing, these form strong contrasts with the highly reflective cirrus cloud ice.

The final two conclusions pertain to the accumulation of selected CLARREO-PF channels using the iterative selection method. The extractable independent information content increases with the increasing number of CLARREO-PF channels that are added to the retrieval system. However, with each channel added to the retrieval selection, the independent information content of all the remaining unselected channels decreases. A similar conclusion was found in a study of the information content of a potential reflective solar hyperspectral retrieval of water clouds properties [10] and with a study of MODIS channels for ice cloud retrievals [7], [16]. Thus, while it is desirable to use all the channels for a cirrus

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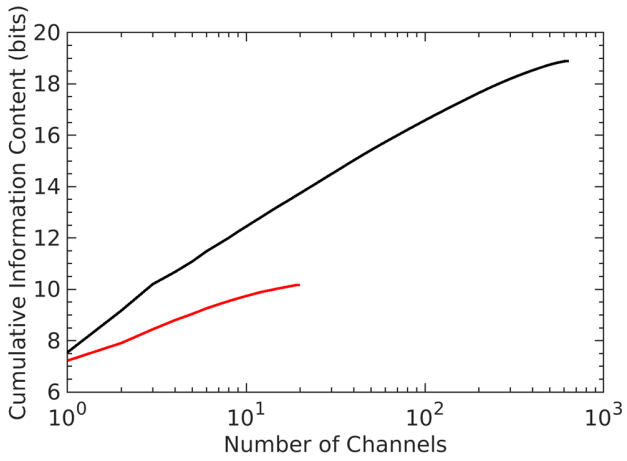


Fig. 11. Comparison of the cumulative information content of two synthetic instruments, a model of CLARREO-PF cumulative information content (black), and an imagined multispectral instrument (red) with MODIS SRF and a flat 5% instrument uncertainty.

cloud optical and microphysical property retrieval, the above two conclusions hint that it is acceptable, but not ideal, to choose those channels that are most important based on the information content.

The vast majority of ice cloud τ_{Cloud} and D_{eff} retrieval algorithms are dual-band algorithms similar to Nakajima and King [9]. An example of this is MODIS collection six retrieval [8], which has been used as the base case for retrieval validations. Such algorithms are based on conservative scattering spectral channels being sensitive to optical depth and not effective particle size, while the ice absorption channels are sensitive to effective particle size. The dual band conservative scattering channels are typically chosen in atmospheric window regions, such as the 0.86 μm channel for the MODIS retrieval [8]. While we found that the water vapor channels are more informative than window channels, it should also be noted that window region channels also inject uncertainty due to reflection off the underlying surface. The use of a water vapor channel limits the surface reflection by absorption below the cloud reducing the additional uncertainty. The selection of water vapor channels as highly informative for optical depth agrees with and supports a series of papers investigating the use of AVIRIS channels around 1.38 and 1.88 μm for identification of cirrus clouds [55], retrieval of cirrus cloud optical depth [56], [57], [58], and the addition to the MODIS of a 1.38 μm channel for cirrus cloud identification [59]. However, there is some literature suggesting that using the AVIRIS 1.38 μm water vapor band does introduce some uncertainty due to sensitivity to surface reflection [58], [60]. These papers suggest using either a 1.88 μm channel instead or paring the 1.38 μm channel with a 1.88 μm channel.

As stated in the introduction, an advantage of using the information content of a measurement for retrieval system characterization is the scalar value that is independent of the retrieval methodology. This allows for comparison between systems. Here, we use SIC to address the added value of the synthetic CLARREO-PF instrument used in this study with respect to a synthetic multispectral instrument that has the same spectral response functions as MODIS [61] within the 350–2300 nm spectral range, a total of 20 channels. It is also assumed that this multispectral instrument has a flat 5% instrument uncertainty. The comparison of the cumulative information content between these two synthetic instruments, as channels are sequentially added, is shown in Fig. 11. The modeled CLARREO-PF cumulative SIC is black, while the multispectral instrument is red. The multispectral instrument necessarily has a lower initial (one channel) SIC due to the lower spectral resolution and greater instrument uncertainty. However, the essential point is the increasing effect of each channel on the final, cumulative, information content. If retrieval systems for both instruments use 20 channels, the one for CLARREO-PF will have a ~4-bit greater information content. If 100 channels are used in the CLARREO-PF retrieval system, the cumulative information content will be ~6 bits greater than the multispectral instrument. If all 640 channels are used in the CLARREO-PF retrieval system, the cumulative information content will be ~9 bits greater than the multispectral instrument.

Finally, a comment needs to be made about high-speed radiative transfer models with respect to the results of this study. Such models include the principal component radiative transfer model for the solar spectral region (PCRTM-Solar; [62]), a superfast model developed as a fast physical reflective solar radiative transfer model for hyperspectral calculations with scattering in the ultraviolet, visible, and near-infrared. Such models are still a maturing field, so the results from our work are useful for many research groups. However, recent literature shows that they offer a different method for dealing with the computational expense in forward model calculations, such as a possible CLARREO-PF cirrus cloud optical and microphysical property retrieval. While PCRTM-Solar has not been used thus far in retrievals, the longwave version, PCRTM [63], has been successfully used in optimal estimation retrievals of water clouds properties [64], [65].

ACKNOWLEDGMENT

This article is based on Section II of the corresponding author's Ph.D. thesis [66] under the supervision of Dr. Ping Yang. Portions of this research were conducted with the advanced computing resources provided by Texas A&M High Performance Research Computing.

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